

PEER REVIEW · BUILD-AGENT HANDOFF · 2026-09-16

TeamsConnector — audio join MCP

https://github.com/SimonBarnett/TeamsConnector
HEAD 00ed1687314649aed2570c46b845ad5fd6bfbf6ea6 (P1 roster / policy_missing / cancelMedia / production secrets)
Claimed: spec v1.2 + OpenAPI 1.5.0. Path A: service-hosted playPrompt + Track A transcripts.

Code is no longer the silent-WAV bot. Worker refuses silence, requires Azure Speech + PUBLIC_BASE_URL, and posts playPrompt. That is not the same as “a human in Teams heard Haitch say the phrase.” docs/spike-track-b.md table is empty. README disagrees with itself on default mode. Disposition: ACCEPT as Path A scaffold. REJECT as tenant-audible MVP until P0-LIVE is a signed row.

0. Verdict

npm test + fixture join/speak will stay green forever on loopback. That path sets audibleInTeams=false. The product promise is attendees hear the bot and the bot returns grounded notes. Neither has a dated tenant log in this repo. Do not ship Hours, camera-tile, or auto plane=media.

Layer	Code	Live?
MCP tools (7) + envelope	Present, tests	Fixture only
Track A resolve + transcripts + VTT	GraphRestClient	Doctor can probe meetings; no transcript log
policy_missing vs 404	GraphHttpError + doctor \$top=1	Needs tenant 403 capture
createCall service-hosted	GraphJoin.CreateCallAsync	Spike table blank
playPrompt of Speech WAV	Program /play + IsSilentPcm refuse	Needs PUBLIC_BASE_URL Graph can GET
cancelMediaProcessing	Wired; mute() does not call it	Unproven
In-memory store	DATABASE_URL fails /ready on purpose	Sessions die on restart
Hours / Calendar workflows	Hidden unless WORKFLOW_TRIGGER=1	Keep hidden

1. Do not regress

- Refuse silent WAV (AzureTts.IsSilentPcm). No empty mediaUri playPrompt.
- healthy=false unless Graph AND Speech keys are set. Loopback is not “media plane up.”
- /play 409 until call established. /cancel calls cancelMediaProcessing, not cancelled:[].
- Production hides fixture Join/Speak HTML. 32-byte ARTIFACT_ENCRYPTION_KEY required.
- No Calls.AccessMedia.All in admin-install. Attendance errors are not swallowed as [].
- listTranscripts 404 → [] is OK for missing transcripts; policy 403 must stay policy_missing.

2. Findings — P0

P0-LIVE Spike table is empty — no audible tenant proof

P0

Files. docs/spike-track-b.md Results table all blank; exit criteria unchecked

Problem. The freeze document still requires: worker joins, speak text heard (not silence), Track A canHear honesty, five consecutive joins. Zero rows.

Why it matters. A build agent will mark Path A done because /play returns status=playing. Graph accepting playPrompt ≠ humans heard the words.

Fix. Run one scheduled test meeting. Fill attempt 1: Admitted, Heard phrase (exact text), PUBLIC_BASE_URL host Graph fetched. Then 2–5. Sign the Decision line.

Acceptance. docs/spike-track-b.md has five dated rows and [x] on join + heard phrase. Phrase in the table matches speak({text}).

P0-PUB playPrompt URI must be reachable by Graph, not by localhost**P0**

Files. services/media-worker/Program.cs PUBLIC_BASE_URL + GET /prompts/{id}.wav; admin-install CALLBACK_URI

Problem. Graph's cloud fetches medialInfo.uri. If PUBLIC_BASE_URL is http://127.0.0.1:7071, playPrompt “succeeds” and nobody hears anything. /prompts is unauthenticated by design so Graph can pull it.

Why it matters. This is the silent-bot failure mode with extra steps.

Fix. Doctor/worker health must fail if PUBLIC_BASE_URL host is loopback. Document ngrok/tunnel. Optionally one-time token on /prompts/{id}?t=. Log Graph's GET on /prompts.

Acceptance. During P0-LIVE, worker access log shows Graph GET 200 on the wav. Humans hear the phrase.

P0-DOC README lies about default mode**P0**

Files. README.md “Default join is listen” vs “Speaks by default (mode=listen_speak, media plane)” vs plane.ts auto+!wantsSpeak → transcript vs spike-track-b default listen

Problem. Three sources, three stories. An agent will join listen_speak on fixture-loopback and tell the user the room heard it.

Why it matters. Consent and plane_unavailable depend on this.

Fix. Pick one: default mode=listen (notes). listen_speak explicit. Fixture speak returns played_locally / audibleInTeams=false (already in last P1 commit — keep it). Fix README first paragraph and the Speaks-by-default bullet. Align doctor copy: graph-waiting-for-worker text still says createCall is unwired; it is wired.

Acceptance. README, spike-track-b, plane.ts, doctor.ts tell the same default. Fixture speak JSON has audibleInTeams=false.

3. Findings — P1

P1-ROSTER Live roster via attendanceReports is the wrong API mid-call**P1**

Files. packages/graph/src/rest.ts getParticipants

Problem. Attendance reports populate after the meeting ends (or late). During an active join this often returns []. Not a swallow-error bug anymore — it is the wrong collection for a live bot.

Why it matters. Agent thinks the room is empty and skips announce / addressing.

Fix. Document: roster may be empty until the report exists. Prefer call roster from calling callbacks if you need live names. Do not block P0-LIVE on this.

Acceptance. README one sentence: participants[] can be empty mid-call.

P1-MUTE mute() drops local prompt id and does not cancel Graph**P1**

Files. Program.cs /mute vs /cancel

Problem. mute returns cancelled ids without CancelMediaProcessingAsync. Audio keeps playing in Teams.

Why it matters. Barge-in / organizer mute feel broken.

Fix. mute = cancelMediaProcessing + clear active. Same as /cancel without requiring utterancelid.

Acceptance. Unit: mute while playing hits cancelMediaProcessing.

P1-STORE In-memory store + DATABASE_URL tripwire**P1**

Files. README; apps/mcp-host compose

Problem. Documented. Sessions vanish on restart. Fine for demo. A Grok bot that joins then dies mid-meeting cannot leave().

Why it matters. Orphan Graph call until timeout.

Fix. leave() in a process shutdown hook. Or say “single-process demo only” in doctor when DATABASE_URL unset.

Acceptance. Killing npm start after join deletes the Graph call in the next demo, or doctor warns orphan risk.

P1-PROMPT Open /prompts/{id}.wav

P1

Files. Program.cs open paths include /prompts/

Problem. Needed so Graph can GET. Utterance ids that are guessable leak meeting audio.

Why it matters. Low if ids are GUIDs and TTL is short.

Fix. GUID utterance ids + delete after playCompleted + 2 minute TTL.

Acceptance. Prompt missing 60s after playCompleted.

4. Findings — P2

P2-MCP Hand-rolled JSON-RPC MCP

P2

Files. apps/mcp-host/src/server.ts

Problem. Works for tools/list and tools/call. No official SDK, no resources, stdio Content-Length framing only.

Why it matters. Grok MCP clients vary.

Fix. Keep if doctor + one stdio session works against the client you ship. Do not rewrite the host this sprint.

Acceptance. tools/list returns 7 tools (workflows off).

P2-404T listTranscripts 404 → []

P2

Files. rest.ts listTranscripts

Problem. Hides a policy 404 if GraphHttpError does not classify it. You already map some 404s to policy_missing on single-resource reads.

Why it matters. canHear=false forever on a meeting that has transcripts.

Fix. Reuse GraphHttpError on this path; collection 403/policy 404 must throw.

Acceptance. Unit: 403 on transcripts → policy_missing, not empty list.

5. Ordered work for the build agent

- **Step 0.** P0-DOC: one default-mode sentence in README + doctor + spike doc.
- **Step 1.** P0-PUB: health.healthy=false if PUBLIC_BASE_URL is loopback. Print the wav URL.
- **Step 2.** P0-LIVE attempt 1 on a scheduled test meeting. Fill spike table row 1.
- **Step 3.** Attempts 2–5. Sign Path A.
- **Step 4.** P1-MUTE. P1-PROMPT TTL.
- **Forbidden:** Calls.AccessMedia.All, Hours auto-post, flipping plane=auto to media before signed spike, claiming fixture speak is audible, ACS rewrite.

6. Review metadata

Field	Value
Repo	https://github.com/SimonBarnett/TeamsConnector
Commit	00ed1687314649aed2570c46b845ad5fd6bfbea6
Prior P1 commit intent	roster, policy_missing, cancel, played_locally, hide workflows, production secrets — present in tree
Not executed here	No Azure tenant join, no playPrompt against Graph
Disposition	REJECT audible MVP. ACCEPT Path A code freeze if P0-DOC is fixed and P0-LIVE is the only remaining ticket.

Next PR title: “P0-LIVE spike row 1 + README default=listen”. Not a new media stack.